ENGINEERING SOCIETY · TECHNICAL TEAM

Idea & Requirements Document

The idea, why it matters, and the precise use cases and workflows that govern what we build.

Governing document, version 1.0

Prepared by Hisham Alhinai, Technical Team Lead

Approved by the Board and Technical Team, August 30, 2026

PART 1 · THE IDEA

Why, how, and what it changes

Where we are today

The society's website today is static. It's a page people scroll through to see who we are, but nothing on it actually works. There is no way to register for an event, no way to see a certificate, no way to browse past projects. Every real action still happens off the website: a Google Form for one event, a WhatsApp message for another, a Drive folder that quietly stops being updated once a committee changes. For example, when Events runs a workshop today, they post a flyer, then share a separate Google Form link in the group chat, and by next semester it's hard to even confirm who attended.

What changes

The website becomes functional instead of a page you just scroll through. When a committee schedules an event, a workshop, or an activity, it gets published directly on the website, and registration happens right there on the same page. No separate form, no separate link to chase down in a chat. For example: Events publishes "Intro to Robotics Workshop, October 12" on the website. A visitor sees it, clicks Register, and is signed up immediately. Everything else we're building (member profiles, attendance, certificates, the archive, the graduate network) connects to that same record.

Advantages, with real examples

- **Members.** A third-year student who attended 12 events over three years can open their profile and see every one of them, plus every certificate earned. Today, that same student would have to dig through years of old email confirmations just to prove they attended.
- **Committees.** Next year's Events team won't rebuild a registration form from scratch. They'll see exactly how many people registered and attended last year's version of the same workshop, and can invite that same list again.
- **The board.** At year-end, the board can pull up a real number, for example "540 members attended 34 events this year," instead of asking every committee to guess from memory.
- **The society long-term.** A graduate from three years ago can still be found in the directory. A current member starting a new project can open last year's magazine archive directly on the platform, instead of asking around for who still has the old Drive link.

PART 2 · REQUIREMENTS

Functional requirements

One line per requirement, each traced to the use case that defines it in full. This is the index. The behavior, steps, and exceptions live in the use case pages that follow, not repeated here.

ID	The system shall...	UC
FR-01	allow a person to create one permanent member profile	UC-01
FR-02	detect duplicate profiles by email or ID and link instead of duplicating	UC-01
FR-03	let a committee publish an event or workshop directly on the website	UC-02
FR-04	let a visitor register for a published event on the same page	UC-02
FR-05	record attendance at check-in against the member and the event	UC-03
FR-06	auto-generate a certificate for every attendee who meets the eligibility rule	UC-04
FR-07	let a committee upload and tag an archive item with project, year, and team	UC-05
FR-08	migrate an existing Drive folder into the archive while keeping it as backup	UC-06
FR-09	update a member's status to Graduate automatically while keeping their history	UC-07
FR-10	let a graduate opt in to a searchable directory and mentorship matching	UC-07

Non-functional requirements

These aren't tied to one use case. They're qualities the whole platform must have, or the core promise (nothing is ever lost again) doesn't actually hold.

Data permanence & integrity

NFR-01. No record is ever hard-deleted. Migrated Drive folders, past events, and old profiles are archived, not removed.

NFR-02. Every change to a record is logged with who made it and when, so history can always be traced back.

Security & privacy

NFR-03. Access is role-based. A committee can only see and edit the data inside its own scope, for example Media cannot edit Events' attendance records.

NFR-04. Personal data (email, ID, phone) is visible only to authorized roles, never exposed in a public view.

NFR-05. A member or graduate can request to see what data is stored about them, or ask to be removed from public directories.

Availability & performance

NFR-06. Event registration stays usable during high-demand moments, such as a popular workshop opening registration to the whole society at once.

NFR-07. Core pages (registration, event browsing, profile) load within a few seconds on a normal mobile connection.

Usability

NFR-08. Registration, event browsing, and certificate access work cleanly on a phone, since most members will use the platform from one, not a laptop.

Backup & recovery

NFR-09. Core data is backed up on a regular schedule, so a technical failure cannot cause permanent loss, the same guarantee we're asking committees to trust us with.

Maintainability

NFR-10. Every module ships with documentation a future module owner can read and understand without asking the person who built it.

PART 2 · REQUIREMENTS

Use case catalog

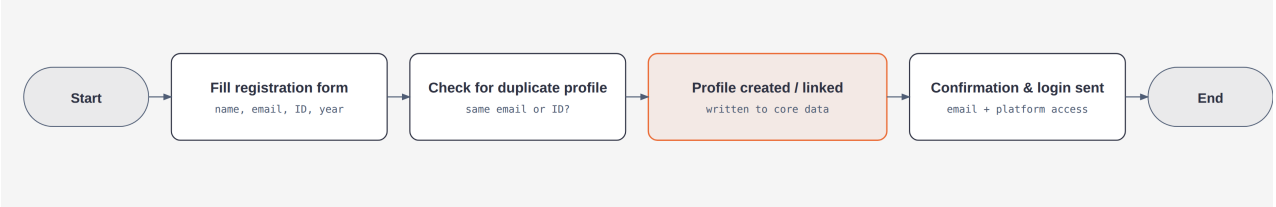
Seven use cases define this phase of the platform, covering every functional requirement above. Each one is detailed on the following pages with its workflow and exact behavior, including what happens when things don't go as planned.

ID	Use case	Primary actor
UC-01	Member registration & profile	Member
UC-02	Website event & workshop announcement + registration	Events/Media committee, Visitor
UC-03	Event attendance tracking	Member, Events committee
UC-04	Digital certificate issuance	Events/Media committee
UC-05	Project / magazine archive upload	Media/Projects committee
UC-06	Drive-to-platform migration	Committee head, Archive owner
UC-07	Graduate network & mentorship matching	Graduating member, Current member

How to read each use case: Actors (who's involved) → Precondition (what must be true before it starts) → Trigger (what kicks it off) → Main flow (the happy path, shown as a diagram and as steps) → Alternate flow (what happens when something's different) → Postcondition (what's true and permanent afterward).

UC-01

Member registration & profile



Happy-path workflow shown above.

Actors	Member, Platform
Precondition	Person has a valid email; society registration is open.
Trigger	Person opens the registration page.

Main flow

1. Member fills the registration form (name, email, ID, year, major).
2. System checks for a duplicate profile (same email or ID).
3. System creates a new member profile in the core data layer.
4. System sends a confirmation email with platform login access.
5. Member logs in, profile is active.

Alternate / exception flow

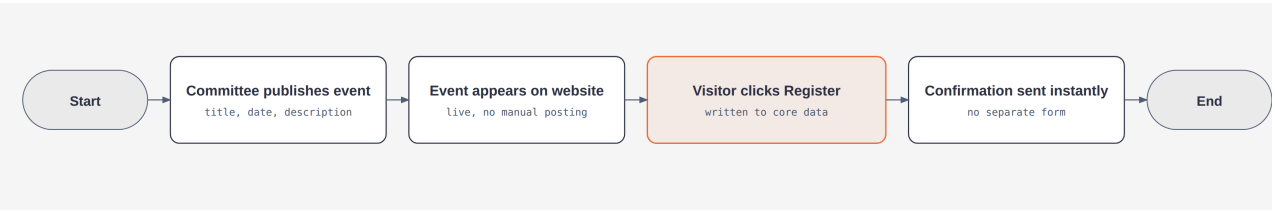
- Duplicate found: system links the registration to the existing profile instead of creating a new one.
- Invalid data: system rejects the form and shows exactly what's missing.

Postcondition

A single, permanent member profile exists and carries forward every future year.

UC-02

Website event & workshop announcement + registration



Happy-path workflow shown above.

Actors	Events/Media committee, Visitor, Platform
Precondition	Committee has an event, activity, or workshop ready to announce.
Trigger	Committee publishes the event on the website.

Main flow

1. Committee publishes event details on the website (title, date, description).
2. The event appears live on the public events page, no manual posting elsewhere.
3. A visitor (member or public) opens the event and clicks Register.
4. Registration is written directly into the platform's core data.
5. Visitor receives confirmation instantly, on the same page.

Alternate / exception flow

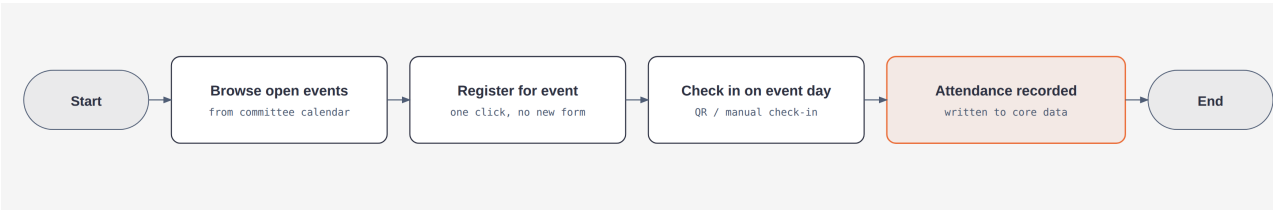
- Approval required: some events may need board or committee review before going live; system holds the event as a draft until approved.
- Event full: website shows a waitlist option instead of open registration.

Postcondition

Every event is publicly discoverable on the website, and registration happens on the platform itself, not a separate Google Form.

UC-03

Event attendance tracking



Happy-path workflow shown above.

Actors	Member, Events committee, Platform
Precondition	Member has registered for a published event (UC-02).
Trigger	Event day arrives and check-in opens.

Main flow

1. Member arrives at the event.
2. Committee checks the member in (QR code or manual check-in).
3. Attendance is written permanently to the member's history and the event's record.
4. Committee sees a live, accurate attendance count during and after the event.

Alternate / exception flow

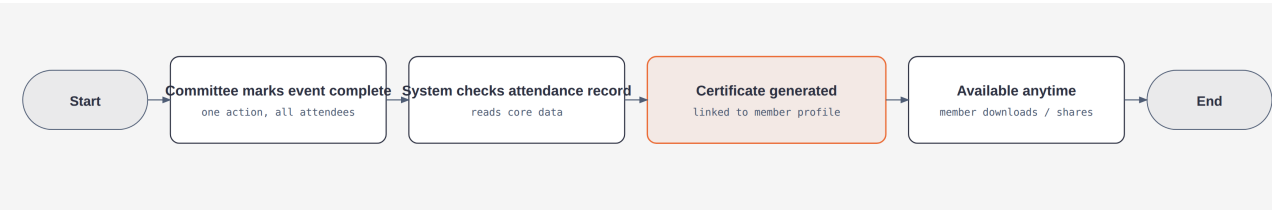
- No-show: record shows "registered, not attended," which affects certificate eligibility (UC-04).
- Walk-in: committee can check in a member who did not pre-register, if space allows.

Postcondition

The event's full attendance record is permanent and tied to every attendee's profile.

UC-04

Digital certificate issuance



Happy-path workflow shown above.

Actors	Events/Media committee, Platform, Member
Precondition	Event has concluded and attendance was recorded (UC-03).
Trigger	Committee marks the event as complete.

Main flow

1. Committee marks the event "complete" in one action.
2. System checks each attendee's record against the eligibility rule (for example, minimum attendance).
3. System auto-generates a certificate for every eligible attendee, linked to their profile.
4. Member can view, download, or share the certificate anytime, with no request needed.

Alternate / exception flow

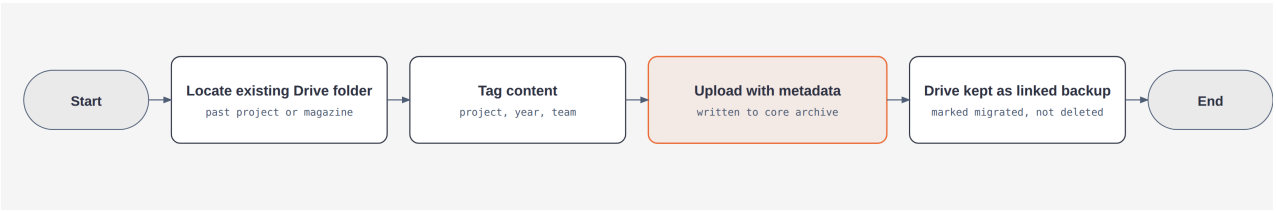
- Not eligible: no certificate is generated automatically; member may request manual committee review.

Postcondition

Every eligible member has a permanent, verifiable digital certificate stored on their profile.

UC-05

Project / magazine archive upload



Happy-path workflow shown above.

Actors	Media/Projects committee, Archive module owner, Platform
Precondition	A finished project, magazine issue, or publication exists.
Trigger	A project or issue is finished and ready to be archived.

Main flow

1. Committee submits the item (file, description, contributors) through the archive module.
2. Item is tagged with project name, year, and team or contributors.
3. System stores it with full metadata in the core archive, linked to contributors' profiles.
4. The item becomes permanently browsable and searchable by the whole society.

Alternate / exception flow

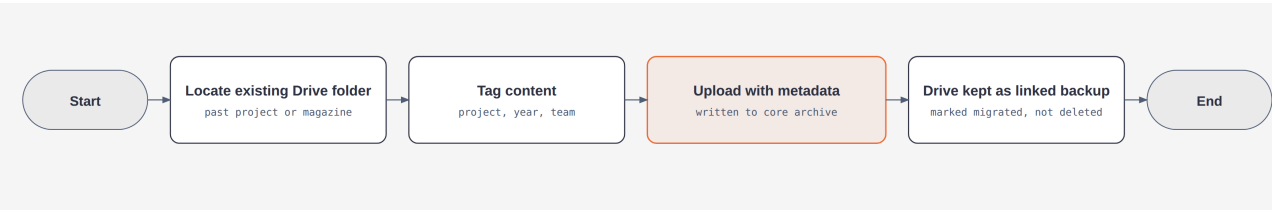
- Missing metadata: system holds the item in a "needs review" state until tagging is complete.

Postcondition

The item is permanently searchable, and appears on every contributing member's history.

UC-06

Drive-to-platform migration



Happy-path workflow shown above.

Actors	Committee head, Archive module owner, Platform
Precondition	A committee still has past years' work stored on Google Drive.
Trigger	The technical team or committee begins migrating a specific Drive folder.

Main flow

1. Committee identifies which Drive folder(s) to migrate first, usually the most-used or most at risk.
2. Archive module owner reviews the contents and tags each item (project, year, team, per UC-05).
3. Items are uploaded into the platform's archive with metadata.
4. The original Drive folder is marked "migrated" and kept as a linked backup, never deleted.

Alternate / exception flow

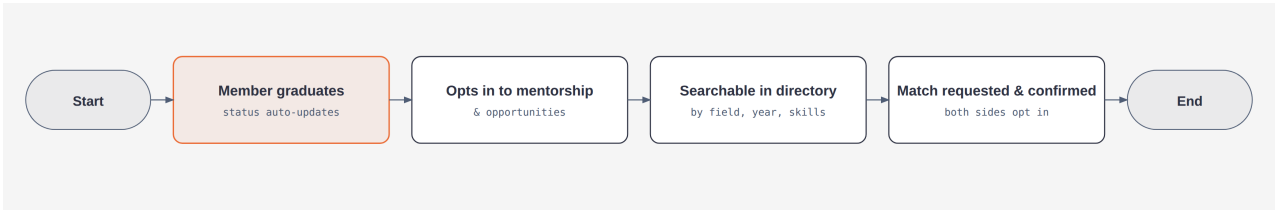
- Incomplete or lost files: missing items are logged as a "known gap," not silently skipped, so the record stays honest about what couldn't be recovered.

Postcondition

Migrated content is permanent and searchable on the platform. The Drive remains only as a backup reference.

UC-07

Graduate network & mentorship matching



Happy-path workflow shown above.

Actors	Graduating member, Current member, Graduate network committee, Platform
Precondition	A member's profile exists and they are completing their studies.
Trigger	Member's status changes to "graduate."

Main flow

1. Member's profile status updates to Graduate. Full history (events, projects, certificates) stays intact.
2. Graduate opts in to appear in the directory and mentorship program.
3. Current members or the committee search the directory by field, graduation year, or skills.
4. A current member requests a mentorship connection or opportunity referral.
5. Both sides confirm, and the connection is recorded.

Alternate / exception flow

- Opt-out: graduate's historical record remains intact but is not publicly searchable.

Postcondition

A living, searchable graduate network exists, and mentorship connections are tracked, not lost in chats.

How the phases work

The phases below are a timeline, but not a calendar. They are ordered by dependency: Phase 2 cannot start until Phase 1 is actually finished and working, however long that honestly takes. We do not move to the next phase on a fixed date. We move when the current phase is done and signed off.



For example, Phase 2 (attendance and certificates, UC-03 and UC-04) only makes sense once Phase 1 is live, because you need real registered members and real events before you can track who attended or issue a certificate. Phase 4 (the graduate network, UC-07) needs member profiles with real history in them, or there is nothing to show a graduate.

The six phases, explained

Phase 0, Foundation. Nothing here is visible to members yet. We set up the data model, the team standards, and who owns what. Every later phase depends on this being done right the first time.

Phase 1, Registration. The first real feature members and committees touch. Members create one permanent profile, and events, activities, and workshops get announced and registered for directly on the website (UC-01, UC-02).

Phase 2, Attendance & certificates. Once events are actually running through the platform, we add attendance check-in and automatic certificate generation for members who qualify, replacing manual certificate creation (UC-03, UC-04).

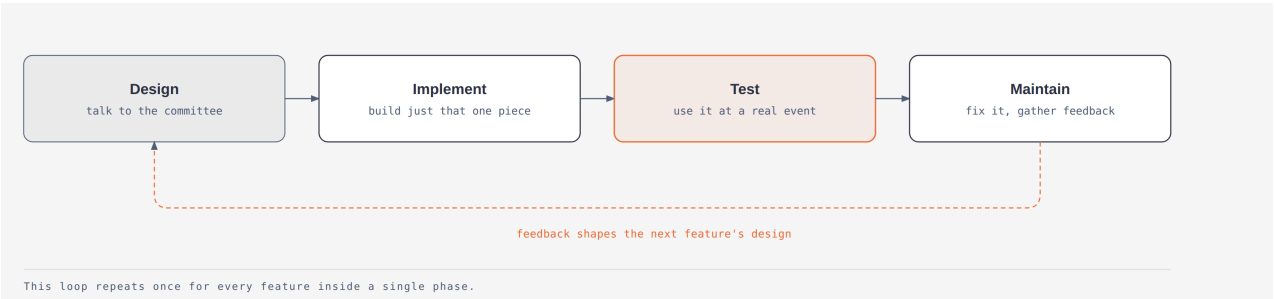
Phase 3, Institutional archive. The project and magazine archive moves onto the platform, including migrating existing Drive folders, so old work becomes searchable instead of scattered across chats and links (UC-05, UC-06).

Phase 4, Graduate network. Once member histories are established, graduating members can stay in a searchable directory, opt into mentorship, and connect with current members for opportunities (UC-07).

Phase 5, AI & auto-reporting. With real data built up from every earlier phase, we add automated reporting for the board and AI-assisted tools on top of that data, instead of manual number-crunching at year-end.

How we build inside each phase

Within a phase, we do not try to plan every feature perfectly upfront and disappear for months. We repeat a short loop, once for every feature.



For example, inside Phase 1 this loop runs once for member registration (UC-01), then again for the website event and registration flow (UC-02). We design with the committee, build only that piece, test it on a real event, then fold what we learned into the design of the next feature. A phase is only marked done once every feature inside it has been through this loop and is actually in use.

Phase mapping

Every use case belongs to one module and one phase. This is what stops anyone from building a feature out of order.

Use case	Module	Phase
UC-01 Member registration & profile	Members	**Phase 1**
UC-02 Website event & workshop announcement + registration	Registration	**Phase 1**
UC-03 Event attendance tracking	Attendance	**Phase 2**
UC-04 Digital certificate issuance	Certificates	**Phase 2**
UC-05 Project / magazine archive upload	Archive	**Phase 3**
UC-06 Drive-to-platform migration	Archive	**Phase 3**
UC-07 Graduate network & mentorship matching	Network	**Phase 4**

GOVERNANCE

How this document stays alive

This document is the shared source of truth for what we're building this phase. Every module owner builds against these use cases, not against a verbal description or a chat message.

- Any change to a use case is written here, not just discussed. If it isn't written, it isn't approved.
- New use cases follow the same format: actors, precondition, trigger, main flow, alternate flow, postcondition.
- This document is versioned. Significant changes bump the version number at the top and get a one-line changelog note.
- Each use case maps directly to a module in the platform architecture and a step in the Phase 0/1 roadmap.

Version log

Version	Date	Approved by	Notes
1.0	Aug 30, 2026	Board + Technical Team	Initial baseline approved. Building begins against this version.